YANG, Yan

[Google Scholar](#) · [GitHub](#) · [LinkedIn](#)

Degrees

Aug 2017 – May 2019 M.Sc. Georgetown University (Washington, DC)

Algorithms | Computer Hardware & System Architecture | Comp. Corpus Linguistics | Empirical Methods in NLP | Stat. Machine Translation | Adv Semantic Representation | Automatic Reasoning | Machine Learning | Deep Reinforcement Learning | Independent Study: Advanced Analytics | Statistical Machine Learning (Ph.D. seminar, approved by instructor) ***Supervisor: Prof. Grace Hui Yang***

Sep 2002 – Jun 2006 B.Eng. Southeast University (Nanjing, China)

* *(2003)* Course Scholarship for Linear Algebra and Space Analytic Geometry*

* *Admitted at age 16*

Linear Algebra & Space Analytic Geometry (scholarship endowed) | Advanced Mathematics | Probability & Statistics | Numerical Computing & Modeling | Digital Image Processing | Speech Signal Processing | Signals & *Linear System | Electromagnetic Field & Waves | Digital Systems & Course Design | Electronic Circuits* & *Comprehensive Experiment | Microcomputer Systems & Interfaces | Fundamentals of Circuit | Analog Electronic Circuits | Modern Psychology | Modern Biology | College Physics | Music Theory | Practice of MATLAB* | etc.

Other education and expertise

(Mar 2026 - now) *The Principles of Quantum Mechanics*, P.A.M.Dirac (§1-20 / Ch 1-3; now on Ch 4) self-study

(Oct 2025) *Autumn School on fMRI (EEG module cancelled)*, IBRO & New York University, Abu Dhabi
• Exposure to experimental design and analysis workflows (Python-based), fMRI principles (HRF, preprocessing), and notebook-based modelling pipelines (e.g., GLM and second-order models).

(Nov 2024 – May 2025) *Elements of Information Theory*, Tomas Cover (fully grasped Ch 1-8 and 14; touched on Ch 11, 13, and 15) self-study

(Aug – Sep 2024) MIT 18.102 *Introduction to Functional Analysis* (went over Lectures 1-8 and 14 Basic Hilbert Space Theory), self-study

(Sep 2023 - Nov 2024) Ph.D. Fellow, at KAUST (Thuwal, Saudi Arabia)[1]

(Apr – May 2023) *Probabilistic Graphical Model 2* (94/100) & *Probabilistic Graphical Model 1* (100/100) on Coursera (Stanford University online)

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[1] *I withdrew from the PhD program after a period of intense gaslighting and repeated dismissal of my scientific concerns. I had raised one early concern regarding validation issues in the supervisor's now-published work, but after being told not to pursue it, I became increasingly confused. Subsequent questions about other aspects of the research directions were also met with dismissal or gaslighting. This created severe inner conflict that left me unable to think clearly or continue productive research. The group subsequently excluded me from further research, citing insufficient output as the reason. The supervisor's published work is currently under journal-level review for research integrity issues (see PubPeer: https://pubpeer.com/publications/7893CE90BA51FEEDCEF83F23B88FFA). I continued research activities during and after this period; related work is documented on Zenodo (https://zenodo.org/records/19397678).*

(May – Jun 2022) *Computational Neuroscience* (99.57/100) on Coursera (University of Washington)

(Jun 2022) *Human Research (Social Behaviour Research Investigators and Key Personnel Group, Basic)* – CITI Program & University of Nevada, Reno

(Jan 2021 – Aug 2022) Ph.D. studies, at University of Nevada, Reno (all courses done)

(May – Jun 2021) *Convolutional Neural Networks* (100/100) on Coursera (Deeplearning.AI)

(Jan – Mar 2021) *Mathematical Biostatistics Boot Camp I* (100/100) on Coursera (Johns Hopkins University)

(Jan 2021) Training: *GRAD 701S – College Teaching* @ University of Nevada, Reno

(Jan 2020) *Certified Generalist Software Engineer* (3% interviewees) – TripleByte

(Jun – Aug 2020) *Genomics: Decoding the Universal Language of Life* (unlicensed) on Coursera (UIUC)

(Jun – Aug 2020) *Finding Hidden Messages in DNA* (unlicensed) on Coursera (UC San Diego)

(July 2017) *Gender Study* - Harvard Summer School (writing A+; other: A) in Beijing (by Harvard University Director of Undergraduate Studies, Prof. Caroline Light)

Research output

Non-refereed and peer-reviewed scientific articles:

<cognitive state modelling>

• Yang, Y. **(2026)**. From Satiation to Allostasis: Toward a Resource-Based Framework for Reversal Triggers in Reversal Theory. *Zenodo*. https://zenodo.org/records/19437171

• Yang, Y. **(2025)**. Modeling Limits in Phenomenological Contexts – a Statement on *Improving Searcher Struggle Detection via the Reversal Theory*. *ResearchGate* Preprint. dx.doi.org/10.13140/RG.2.2.28606.80961

• Jiyun, L., Yang, Y., Nayak, V., & Yang, G.H. **(2024)**. Improving Searcher Struggle Detection via the Reversal Theory. *Discover Computing* 27, 51. Springer Nature. doi.org/10.1007/s10791-024-09492-z

<generative machine learning>

• Yang, Y. **(2026)**. EM-style Framework for Task-Driven Differentiable Synthesizer Optimization: Rationale, Design, and Backpropagation through Reaction–Diffusion System. *Zenodo*. https://zenodo.org/records/19397678

• Yang, Y., Bebis, G., & Nicolescu, M. **(2022)**. Two-Step Data Augmentation for Masked Face Detection and Recognition: Turning Fake Masks to Real. In Proceedings of *the 2nd International Conference on Image Processing and Vision Engineering – IMPROVE* (pp. 126-134). SciTePress. dx.doi.org/10.5220/0011037900003209

Methods, software, infrastructures, materials, guides and tools developed

(2025) Investigated the feasibility of **gradient-based parameter learning** in **reaction–diffusion systems** (Gray–Scott model), focusing on **backpropagation** via **time-stepped PDE simulations**. After repeated failure modes, developed a structured **experimental codebase** that **enables systematic probing** of gradient behavior, numerical mechanisms, and optimizer dynamics via truncated BPTT, adaptive learning rates, embedded simulation animation, and loss landscape plotting. This work supports iterative refinement, including using surrogates with better loss landscapes, **toward a viable differentiable implementation**. Code is at: https://github.com/Yan-Yang-bot/bp2renderer. *Independent Research @ self*

(2021 ~ 2022) Led the prototyping of the Neural Network **embedding visualization system** and took charge of the **system design** of its remodelling, developing it into a **key "human-in-the-loop" step** of the Cluster-Based Sorting product. *R&D Intern / R&D Engineer @ Deepcell Inc.*

(2021) Developed **installation guides** for the local-running prototype version of CNN embedding visualization tool. *R&D Intern @ Deepcell, Inc.*

(2018) Revamped search engine's **VR interface** (**hand model formulation** and connection) for **behavioral data collection** and provided **web technique** support for its SIGIR conference demonstration, work acknowledged by the corresponding author as authorship-worthy. *Research Assistant @ Georgetown University*

(2015) Led the development of a weather report app with **dynamic charts** for precipitation and other weather metrics (AngularJS, Flask, NodeJS, AliCloud, Nginx, MySQL). **Data API** monitoring. **Precipitation interpolation**. *Full Stack Engineer @ ColorfulClouds Tech Co. Ltd*

Scientific and societal impact

(Sep – Oct 2025) Active Participant @ Models of Consciousness Conference 6

Contributed to meta-discussions on consciousness modeling by introducing **Gödel's incompleteness theorem**.

(Apr 2025 – now) LinkedIn content creator

Visualisations & essays for **mathematical** understanding, audiences included academic professors, researchers, startup co-founders and top company engineers:

- **Animation**: Dirac delta approximation via cosine summation
- Visual diagram of matrix classifications in **advanced linear algebra**
- Short essay on the development of **non-Euclidean geometry**, with a visual timeline

(Sep 2022) Active Participant @ Models of Consciousness Conference 3

Raised discussion topics on **timescales of psychological phenomena** and related **data collection**; engaged in post-talk exchanges with faculty from CMU (CS, **subconscious models**) and Stanford (philosophy of mind, **qualia**).

(Apr 2022) Presenter @ IMPROVE (online due to COVID-19)

First-author paper *Two-Step Data Augmentation for Masked Face Detection and Recognition*.

Language skills

Mandarin – Native. **English** – Fluent C2.
6 years in the US; 7 years science/tech work in English; Graduate schools in English; Translated 19th century Old English in a full-length manuscript.
Certificates: Dec 2022, *Duolingo* 135; Nov 2016, *TOEFL* 105.

French – Basic A1.
Greetings; Recognize simple words, sentence structures, basic tenses in general books/packages/billboards.

Traditional Chinese – Basic A2.
Basic Duolingo & YouTube courses done; Simple talk with natives; Read beyond its Mandarin writing similarity.

Chronological work experience[2]

Jul – Nov 2025   Reviewer @ IEEE Transactions on Affective Computing

Reviewed machine-learning manuscripts:
• Identified deeper analytical questions necessary for interpreting empirical results.
• Corrected a serious conceptual misuse of cross-disciplinary terminology.
• Helped strengthen the coverage and selection of related work.
• Proposed structural revisions to the writing.

Oct 2025 – Current   Independent Researcher

• Independently continued KAUST-era research on task-driven differentiable renderer optimization, shifting focus to direct backpropagation through unrolled Gray-Scott time steps. Developed a comprehensive experimental codebase for systematic diagnosis of gradient behavior, including truncated BPTT, adaptive learning rates, alternative loss functions, loss landscape visualization, and embedded simulation animation tools. Evaluated intermediate supervision, implicit differentiation, PINN, and surrogate model approaches as potential next steps.
• Independently extended prior work on Reversal Theory by exploring theoretical limits of its application in computational modeling and proposing linkages between reversal triggers and Barrett's predictive allostasis (resource budgeting) framework, as well as connections between measurement disturbance in phenomenological states and quantum measurement principles.
• Participated in BAMΞ workshop on measurement (online), Models of Consciousness Conference 6 (Sapporo, Japan), and IBRO Autumn School on fMRI (NYU Abu Dhabi).

Nov 2024 – Sep 2025 <Career Gap> *Freelance work, theoretical consolidation, and limited independent research; activities included journal reviewing, independent writing, mathematical visualization works, theoretical consolidation, and doctoral application preparation; included a medical recovery phase.*

Jul 2024 – Jan 2025   Reviewer @ IEEE Transactions on Neural Networks and Learning Systems

Sep 2023 – Nov 2024   Research Fellow @ KAUST (Thuwal, Saudi Arabia)

• Explored methods to learn reaction-diffusion systems (i.e., PDE parameters) from data; the underlying mathematics connects to neuroscience modeling, and in this project was applied to automate synthesis of diseased leaf images for agricultural classification models.
• Proposed ideas on Arabic Speech Recognition, including acoustic source separation and data augmentation in the spectrogram.

May – Aug 2023 <Career Gap> *Road trip across half the US and personal life-planning milestone after a layoff and concurrently obtaining a Ph.D. offer.*

Oct 2022 – May 2023   Machine Learning R&D Engineer @ Deepcell Inc. (Mountain View, United States)

• Led the system design of the remodeled Neural Network embedding visualization system, which is to be used seamlessly in the Cluster-Based Sorting product. When the imaging phase of an imaging+sorting run is done, we use this design to pause the run and start a data exploration session to select sorting criteria.
• My system design covered decisions on data storage media, database schemas, API specs,

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[2] *This section outlines all roles, academic, industrial, and volunteer, intentionally arranged chronologically to reflect the evolution of my research and professional interests*

serialization/deserialization options, caching, parallelizing large calculations, and others for incremental developing plans and long-term goals.

Sep 2022 <Career Gap> *One-month gap between OPT/H-1B work permits (not allowed to work by the US Immigration Bureau); attended Models of Consciousness Conference 3.*

Jun – Aug 2022 Volunteer Research Assistant @ Visual Perception Lab - Neuroscience, University of Nevada, Reno (Reno, United States)

• Trained for BrainVoyager usage and human research ethics.
• Helped with software and package installations and documented an installation guide.
• Read papers on Representational Similarity Analysis, Multivoxel Pattern Analysis, human brain's face/object perception, fMRI, etc., & discussed research ideas with other lab members based on their existing code and data.

Jan – May 2022 Teaching Assistant @ University of Nevada, Reno (Reno, United States)

Recitation classes, grading, office hours, & exam proctorship, MATH 127, Precalculus II, including preparation of structured recitation slides and class exercises.

May 2021 – Sep 2022 R&D Intern @ Deepcell Inc. (Mountain View, United States)

• Trained GAN models, supplementing data for cell image classifiers.
• Led the development of a comprehensive CNN embedding layer visualization tool, used company-wide by data and bioinformatic scientists for clustering, correlation analysis, etc., and greatly improved their productivity.

Sep 2021 – May 2022 Researcher @ University of Nevada, Reno (Reno, United States)

• Engaged in a variety of course research projects, such as HPC log data analytics, medical data segmentation and classification, and adversarial learning used for data augmentation.
• Published Two-Step Data Augmentation for Masked Face Detection and Recognition: Turning Fake Masks to Real.

Jan – May 2021 Teaching & Research Assistant @ University of Nevada, Reno (Reno, United States)[3]

Grading and office hours, CS 320, Data Structure (C++).

Jun 2020 – Jan 2021 Member of Technical Staff @ Pure Storage (Bellevue, United States)

• Implemented feature: import/restore volume snapshots from company disks into Kubernetes.
• Enhancement/bug fixing: multipath device problem, upgrading bash scripts to Python3, etc.
• Led the GitHub release process of Pure Service Orchestrator v 6.0.3.

Jun 2019 – May 2020 <Career Gap> *Completed research postponed due to making up coursework after an in-semester medical leave; began job search, with placement delayed by the onset of COVID-19.*

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[3] *Research assistant funding was cut abruptly; subsequently pursued an industry internship and a teaching role to bridge the gap while continuing to seek research opportunities.*

Jan 2018 – Sep 2019 Research Assistant @ Georgetown University (Washington, United States)[4]

@InfoSense Lab: 1) Translated psychological reversal theory into statistical languages to analyze search engine user behaviors (search logs) and improve user struggle detection; 2) Revamped the search engine's VR interface for behavioral data collection, and provided web technique support for its SIGIR '18 conference demonstration, work acknowledged by the corresponding author as authorship-worthy.
@Massive Data Institute: "Victims of Violence" tweets analytics: occupational analysis and database management.

Sep 2016 – Jul 2017 <Career Gap> *Language **training** and graduate-school applications; attended **Harvard Summer School** (Gender Study).*

Jun – Aug 2016 Web Development Engineer & Mentor @ DataMesh (Beijing, China)

• Led the development of a news publishing system implemented as a React-Redux app;
• Solved Apache deployment conflicts among static files, React router hack, and Python service;
• Mentored new employees on webpage building.

Jan – Jun 2016 Software Engineer @ Delicious Meals Delivery Tech Ltd. (Beijing, China)

Designed and implemented a message push system using WebSocket, AngularJS, and GoLang Micro Service for the meal website, which solved user complaints that pages don't sync the latest meal status.

Jul 2015 – Jan 2016 Full Stack Engineer @ ColorfulClouds Tech Ltd. (Beijing, China)

• Independently took charge of a weather report app with dynamic charts for weather metrics (e.g., precipitation).
• Implemented a compatibility patch for data API v2, providing continued support to v1 clients.
• Automatic periodic collection of API health data into the Redis database.
• Precipitation interpolation using NumPy.

Oct 2014 – Jun 2015 Frontend Developer @ One Lead Interactive Media Advertising Co., Ltd., etc. (Beijing, China)

• Made WeChat minigames for marketing activities: https://github.com/yan-yang-bot/MiniGame-Zillionaire.
• Built webpages for an information management app, using D3.js for data reports and charts.

Sep 2013 – Oct 2014 Web Development Engineer @ EnglishPass100 Education Tech. Ltd., etc. (Beijing, China)

Developed C++-based Windows apps (ID card verifier, Word Solitaire game) with Linux server.

Mar – Sep 2013 <Career Gap> *Career transition from translation to software engineering via **self-study in C++; independently developed a minesweeper game**.*

Jan 2009 – Sep 2013 Translator (EN-CN) @ Self-employed (Beijing, China)

• Translated and organized the book Women in the 19th Century (Margaret Fuller, 1843), ensuring accuracy and consistency across a full-length manuscript.

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[4] *To avoid confusion, during this period, I contributed to two research projects: one resulting in a journal publication five years later (2024), and another involving a VR interface for SIGIR 2018, which the corresponding author recognized as authorship-worthy.*

• Translated and proofread marketing materials, user/developer manuals, TV subtitles, and software/game *content for clients including Yeeyan.org and Grand Strong*.

<Scientific News Translator> from 2009 to 2010

• Published dozens of Chinese translations of science and technology news from media outlets such as Wired.com.

Jan 2008 – Jan 2009 Scientific Secretary @ Tongji University (Shanghai, China)

Jul 2006 – Dec 2008 <Career Gap> *Mental health recovery following sexual misconduct and other incidents in undergraduate; engaged in* ***self-study of psychology and philosophy****. (Details omitted for academic focus.)*

Jul 2006 – Dec 2008 Psychology Peer Supporter (Volunteer) @ Ayaohelp.com (Remote, China)

• Provided ***peer support*** *informed by* ***self-study in psychology and philosophy****, focusing on strategic life-event analysis and knowledge sharing. In addition to practice-oriented interpretive approaches commonly used in counselling contexts, I intentionally covered topics in experimental psychology and general psychology, e.g., cognition, perception, learning, memory, basic principles of research design, measurement, and validity;*

• Reflective ***essays/poems*** *on* ***personal development*** *were* ***highlighted on front page*** *by the site host, a Psychology professor at Beijing Normal University, to engage readers.*